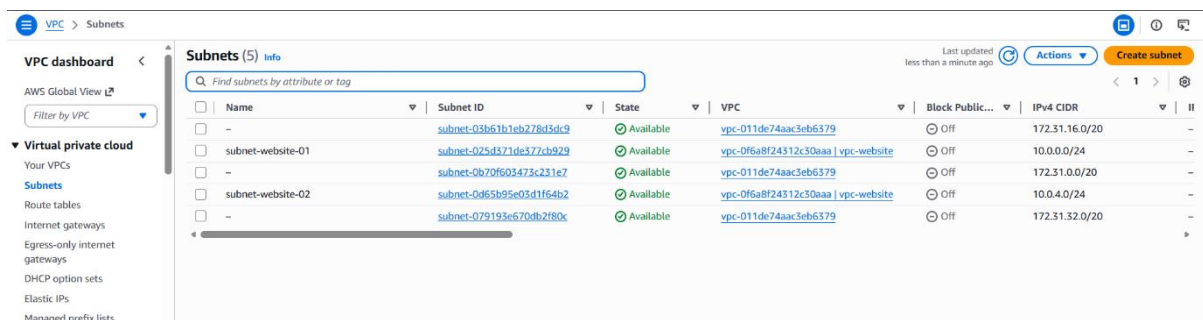
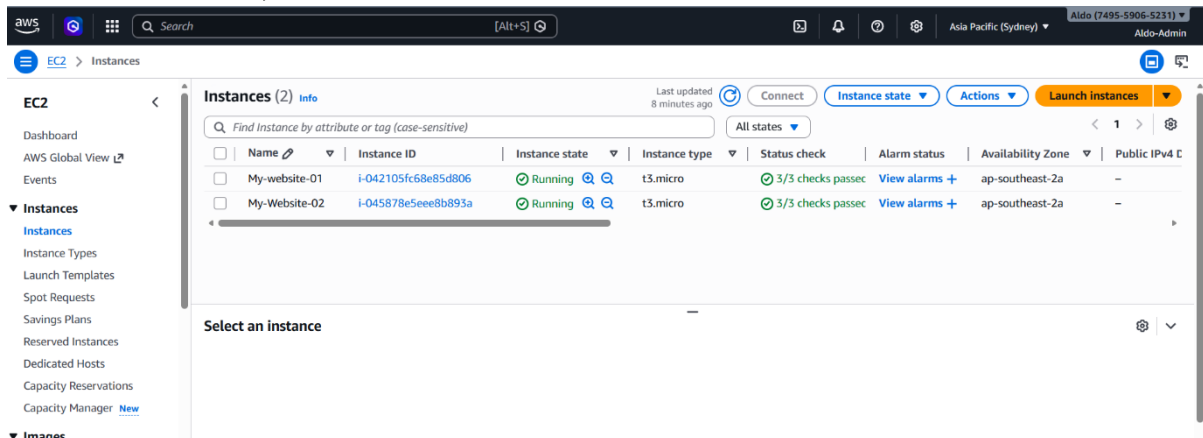
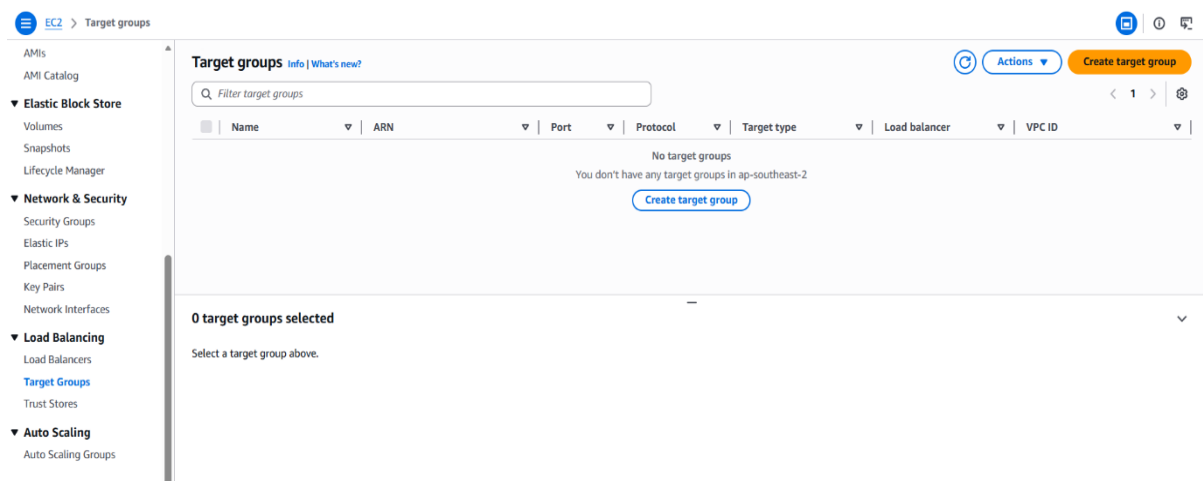


HOW TO CREATE AUTO LOAD BALANCER (ALB)

1. Make sure you have 2 instance. (if you just have one , just create like other one and give different name in server)
2. Make sure you have 2 subnet with different availability zone. (this is useful if you want create load balancer)



3. Then create target group.
4. In aws console , select load balancing and then choose target group.
5. In dashboard target group , click create target group.



6. In this setting , select Instances.

7. Create target group name.

EC2 > Target groups > Create target group

Step 1
● Create target group
○ Step 2 - recommended
○ Register targets
○ Step 3
○ Review and create

Create target group

A target group can be made up of one or more targets. Your load balancer routes requests to the targets in a target group and performs health checks on the targets.

Settings - immutable
Choose a target type and the load balancer and listener will route traffic to your target. These settings can't be modified after target group creation.*

Target type
Indicate what resource type you want to target. Only the selected resource type can be registered to this target group.

☒ **Instances**
Supports load balancing to instances in a VPC. Integrate with Auto Scaling Groups or ECS services for automatic management.
Suitable for: ALB NLB GWLB

☐ **IP addresses**
Supports load balancing to VPC and on-premises resources. Facilitates routing to IP addresses and network interfaces on the same instance. Supports IPv6 targets.
Suitable for: ALB NLB GWLB

☐ **Lambda function**
Supports load balancing to a single Lambda function. ALB required as traffic source.
Suitable for: ALB

☐ **Application Load Balancer**
Allows use of static IP addresses and PrivateLink with an Application Load Balancer. NLB required as traffic source.
Suitable for: NLB

Target group name
Name must be unique per Region per AWS account.
websiteTarget
Accepts: a-z, A-Z, 0-9, and hyphen (-). Can't begin or end with hyphen. 1-32 total characters; Count: 13/32

8. In IP address type , select IPv4

9. Select the VPC that you created before.

10. In Protocol version select HTTP1 , and then click next.

IP address type
Only targets with the indicated IP address type can be registered to this target group.

☒ **IPv4**
Each instance has a default network interface (eth0) that is assigned the primary private IPv4 address. The instance's primary private IPv4 address is the one that will be applied to the target.

☐ **IPv6**
Each instance you register must have an assigned primary IPv6 address. This is configured on the instance's default network interface (eth0). [Learn more](#)

VPC
Select the VPC with the instances that you want to include in the target group. Only VPCs that support the IP address type selected above are available in this list.

vpc-0f6a8f24312c30aaa (vpc-website)
10.0.0.0/16

[Create VPC](#)

Protocol version

☒ **HTTP1**
Send requests to targets using HTTP/1.1. Supported when the request protocol is HTTP/1.1 or HTTP/2.

☐ **HTTP2**
Send requests to targets using HTTP/2. Supported when the request protocol is HTTP/2 or gRPC, but gRPC-specific features are not available.

☐ **gRPC**
Send requests to targets using gRPC. Supported when the request protocol is gRPC.

11. In register target dashboard , select 2 instance your created before, and then click include as pending bellow.

EC2 > Target groups > Create target group

Step 1
○ Create target group
● Register targets
○ Step 3
○ Review and create

Register targets - recommended

This is an optional step to create a target group. However, to ensure that your load balancer routes traffic to this target group you must register your targets.

Available instances (2/2)

Filter instances

☒	Instance ID	Name	State	Security groups	Zone	Private IP
☒	i-045878e5ee8b893a	My-Website-02	Running	MySGwebsite	ap-southeast-2a	10.0.0.249
☒	i-042105fc68e85d806	My-website-01	Running	MySGwebsite	ap-southeast-2a	10.0.0.212

2 selected

Ports for the selected instances
Ports for routing traffic to the selected instances.

80
1-65535 (separate multiple ports with commas)

[Include as pending below](#)

12. In register targets , you will see you have 2 target instance , and then click create target group.

Step 2: Register targets [Edit](#)

Targets (2)

Instance ID	Name	Port	Zone
i-045878e5eee8b893a	My-Website-02	80	ap-southeast-2a
i-042105fc68e85d806	My-website-01	80	ap-southeast-2a

[Cancel](#) [Previous](#) [Create target group](#)

13. Back to aws console, select load balancer.

14. click create load balancer.

15. In Load balancer types , select Application load balancer and then click create.

Load balancer types

Application Load Balancer [Info](#)

Choose an Application Load Balancer when you need a flexible feature set for your applications with HTTP and HTTPS traffic. Operating at the request level, Application Load Balancers provide advanced routing and visibility features targeted at application architectures, including microservices and containers.

[Create](#)

Network Load Balancer [Info](#)

Choose a Network Load Balancer when you need ultra-high performance, TLS offloading at scale, centralized certificate deployment, support for UDP, and static IP addresses for your applications. Operating at the connection level, Network Load Balancers are capable of handling millions of requests per second securely while maintaining ultra-low latencies.

[Create](#)

Gateway Load Balancer [Info](#)

Choose a Gateway Load Balancer when you need to deploy and manage a fleet of third-party virtual appliances that support GENEVE. These appliances enable you to improve security, compliance, and policy controls.

[Create](#)

16. In dashboard basic configuration , fill load balancer name.

17. In scheme select internet facing.

18. In load balancer IP address type select IPv4

Basic configuration

Load balancer name
Name must be unique within your AWS account and can't be changed after the load balancer is created.

A maximum of 32 alphanumeric characters including hyphens are allowed, but the name must not begin or end with a hyphen.

Scheme [Info](#)
Scheme can't be changed after the load balancer is created.

☒ **Internet-facing**
• Serves internet-facing traffic.
• Has public IP addresses.
• DNS name resolves to public IPs.
• Requires a public subnet.

☐ **Internal**
• Serves internal traffic.
• Has private IP addresses.
• DNS name resolves to private IPs.
• Compatible with the IPv4 and Dualstack IP address types.

Load balancer IP address type [Info](#)
Select the front-end IP address type to assign to the load balancer. The VPC and subnets mapped to this load balancer must include the selected IP address types. Public IPv4 addresses have an additional cost.

☒ **IPv4**
Includes only IPv4 addresses.

☐ **Dualstack**
Includes IPv4 and IPv6 addresses.

☐ **Dualstack without public IPv4**
Includes a public IPv6 address, and private IPv4 and IPv6 addresses. Compatible with **Internet-facing** load balancers only.

19. In network mapping , select VPC you already created.

20. And in availability zone and subnet select both subnet. (in here ap-southeast-2a & ap-southeast-2c)

Network mapping [Info](#)
The load balancer routes traffic to targets in the selected subnets, and in accordance with your IP address settings.

VPC [Info](#)
The load balancer will exist and scale within the selected VPC. The selected VPC is also where the load balancer targets must be hosted unless routing to Lambda or on-premises targets, or if using VPC peering. To confirm the VPC for your targets, view [target groups](#).

10.0.0.0/16

[Create VPC](#)

IP pools [Info](#)
You can optionally choose to configure an IPAM pool as the preferred source for your load balancers IP addresses. Create or view Pools in the [Amazon VPC IP Address Manager console](#).

☐ **Use IPAM pool for public IPv4 addresses**
The IPAM pool you choose will be the preferred source of public IPv4 addresses. If the pool is depleted IPv4 addresses will be assigned by AWS.

Availability Zones and subnets [Info](#)
Select at least two Availability Zones and a subnet for each zone. A load balancer node will be placed in each selected zone and will automatically scale in response to traffic. The load balancer routes traffic to targets in the selected Availability Zones only.

☒ **ap-southeast-2a (apsec2-az1)**
Subnet
Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently.

10.0.0.0/24

☒ **ap-southeast-2c (apsec2-az2)**
Subnet
Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently.

10.0.4.0/24

21. Select the security group that you have created earlier.

22. In routing action select forward to target groups.

23. In target group , select target group your created earlier, and then click create load balancer.

Security groups [Info](#)
A security group is a set of firewall rules that control the traffic to your load balancer. Select an existing security group, or you can [create a new security group](#).

Security groups
Select up to 5 security groups

Default action [Info](#)
The default action is used if no other rules apply. Choose the default action for traffic on this listener.

Routing action

☒ **Forward to target groups** ☐ **Redirect to URL** ☐ **Return fixed response**

Forward to target group [Info](#)
Choose a target group and specify routing weight or [create target group](#).

Target group

Target type: Instance, IPv4 | Target stickiness: Off

Weight

0-999

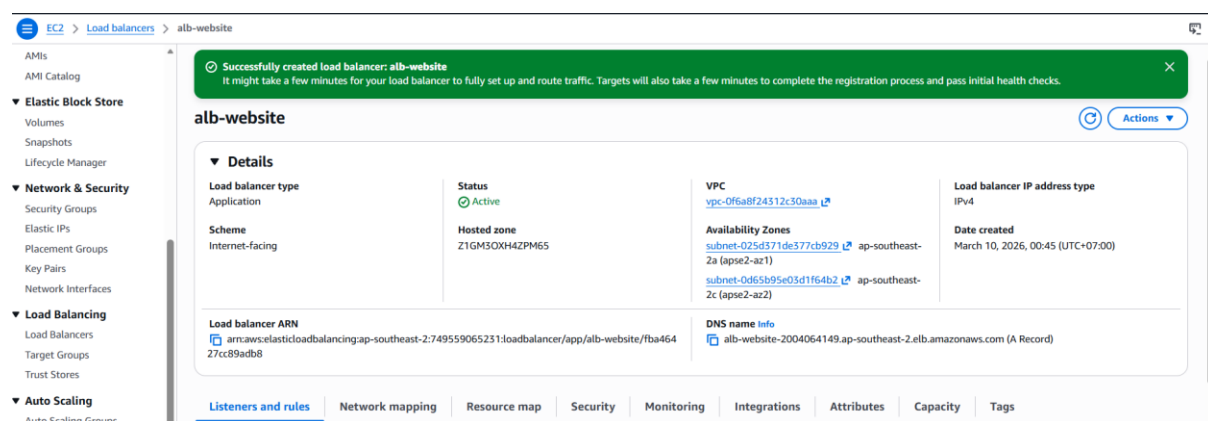
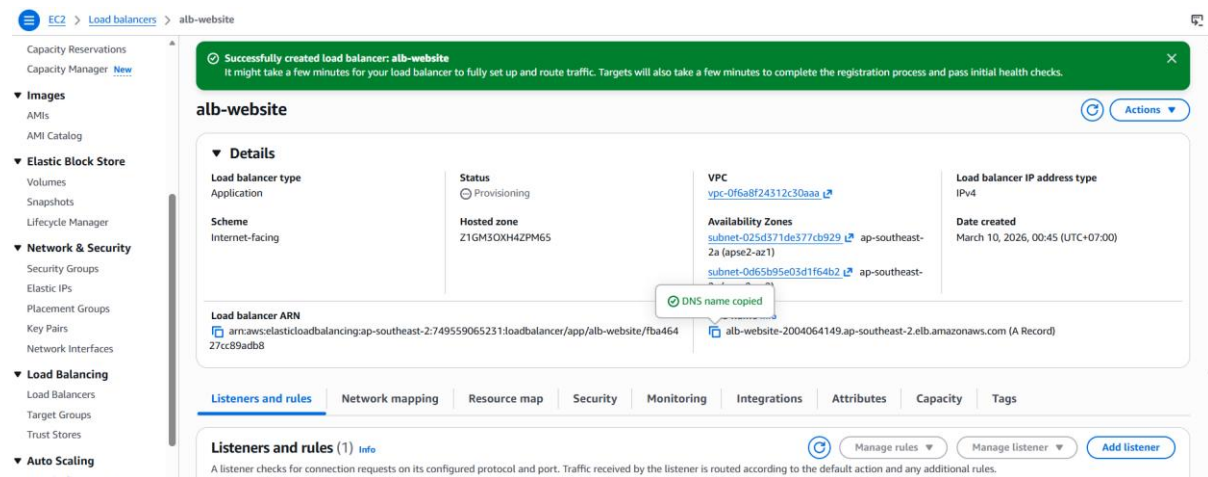
Percent
100%

[+ Add target group](#)
You can add up to 4 more target groups.

24. In here you can see at dashboard details and you can see status still (Provisioning) just wait 2 – 5 minute until status changes to active.

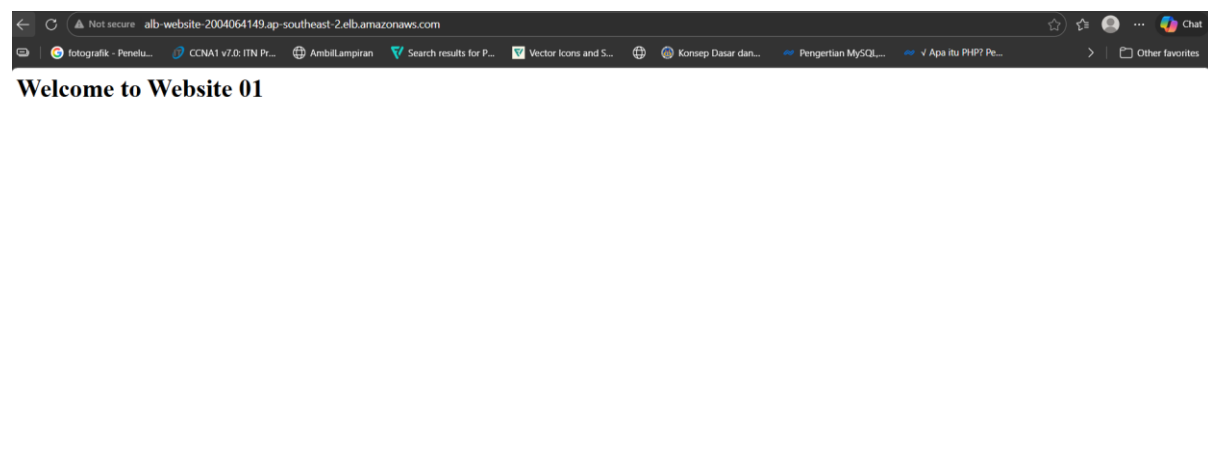
25. After status active , copy DNS name.

26. Open new tab and Paste DNS name.



27. you will be redirected to the website page that you have created.

(For make sure you Auto load balancer work, click refresh several time until you web page change like this image)



Welcome to website 02